

Discrete Random Variable	Continuous Random Variable								
<u>X - DRV</u> $X = x_1, x_2, \dots, x_n$ $x_i \in \mathbb{R}$ <u>Probability Mass function</u> (p.m.f) <table border="1"> <tr> <td>x_i</td> <td>x_1</td> <td>$\dots$</td> <td>x_n</td> </tr> <tr> <td>p_i</td> <td>p_1</td> <td>$\dots$</td> <td>p_n</td> </tr> </table> (i) $p_i \geq 0$ (+ve nos) (ii). $\sum_{i=1}^n p_i = 1$ p_i's are p.m.f $\{x_i, p_i\}_{i=1}^n$ is called prob. distribution of. a DRV X	x_i	x_1	$\dots$	x_n	p_i	p_1	$\dots$	p_n	<u>X - CRV</u> $X = x \in [a, b] = \mathbb{R}_x$ $[a, b] \subseteq \mathbb{R}$ <u>Prob. Density function</u> (p.d.f) $f(x)$ is said p.d.f if (i). $f(x) \geq 0$, (+ve nos) $\forall x \in \mathbb{R}_x$ (ii) $\int_{\mathbb{R}_x} f(x) dx = 1$ $\{x, f(x)\}_{x \in \mathbb{R}_x}$ is called P.D of a CRV X
x_i	x_1	$\dots$	x_n						
p_i	p_1	$\dots$	p_n						

Discrete Random Variable

If X is a random variable (RV) which can take a finite number or countably infinite number of values, X is called a discrete RV.

Probability Function

If X is a discrete RV which can take the values $x_1, x_2, x_3, \dots$ such that $P(X = x_i) = p_i$, then p_i is called the *probability function* or *probability mass function* or *point probability function*, provided p_i ($i = 1, 2, 3, \dots$) satisfy the following conditions:

(i) $p_i \geq 0$, for all i , and

(ii) $\sum_i p_i = 1$

The collection of pairs $\{x_i, p_i\}$, $i = 1, 2, 3, \dots$, is called the *probability distribution* of the RV X , which is sometimes displayed in the form of a table as given below:

$X = x_i$	$P(X = x_i)$
x_1	p_1
x_2	p_2
$\vdots$	$\vdots$
x_r	p_r
$\vdots$	$\vdots$

Continuous Random Variable

If X is an RV which can take all values (i.e., infinite number of values) in an interval, then X is called a *continuous* RV.

Probability Density Function

If X is a continuous RV such that

$$P\left\{x - \frac{1}{2} \leq X \leq x + \frac{1}{2}\right\} = f(x)dx$$

then $f(x)$ is called the *probability density function* (shortly denoted as pdf) of X , provided $f(x)$ satisfies the following conditions:

(i) $f(x) \geq 0$, for all $x \in R_x$ and

(ii) $\int_{R_x} f(x)dx = 1$

Moreover, $P(a \leq X \leq b)$ or $P(a < X < b)$ is defined as

$$P(a \leq X \leq b) = \int_a^b f(x)dx.$$

When X is a continuous RV

$$P(X = a) = P(a \leq X \leq a) = \int_a^a f(x) dx = 0$$

Cumulative Distribution Function (cdf)

If X is an R.V., discrete or continuous, then $P(X \leq x)$ is called the *cumulative distribution function* of X or *distribution function* of X and denoted as $F(x)$.

If X is discrete,

$$F(x) = \sum_{X_j \leq x} p_j$$

If X is continuous,

$$F(x) = P(-\infty < X \leq x) = \int_{-\infty}^x f(x) dx$$

Properties of the cdf $F(x)$

1. $F(x)$ is a non-decreasing function of x , i.e., if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$.
2. $F(-\infty) = 0$ and $F(\infty) = 1$.
3. If X is a discrete R.V. taking values $x_1, x_2, \dots$, where $x_1 < x_2 < x_3 < \dots < x_{i-1} < x_i < \dots$, then $P(X = x_i) = F(x_i) - F(x_{i-1})$.
4. If X is a continuous R.V., then $\frac{d}{dx} F(x) = f(x)$, at all points where $F(x)$ is differentiable.

Special Distributions

Discrete Distributions

1. *Binomial Distribution*
2. *Poisson Distribution*

Continuous Distributions

1. *Normal Distribution*
2. *Exponential Distribution*
3. *Gamma Distribution*
4. *Weibull Distribution*

X - RV

$$E(X) = \frac{\sum_{i=1}^n x_i p_i}{\sum p_i} = \sum_{i=1}^n x_i p_i \quad \text{if } X \text{ is DRV}$$

$$\downarrow$$

Mean

$$\int_{-\infty}^{\infty} x f(x) dx \quad \text{if } X \text{ is CRV}$$

$$E(X^2) = \begin{cases} \sum_{i=1}^n x_i^2 p_i & \text{if } X \text{ is DRV} \\ \int_{-\infty}^{\infty} x^2 f(x) dx & \text{if } X \text{ is CRV} \end{cases}$$

Let $g(x)$ be a function on a RV X .

$$E(g(X)) = \begin{cases} \sum_{i=1}^n g(x_i) \cdot p_i & \text{if } X \text{ is DRV} \\ \int_{-\infty}^{\infty} g(x) f(x) dx & \text{if } X \text{ is CRV} \end{cases}$$

Suppose $g(x) = x$ (Identity function)
 $E(g(X)) = E(X)$

Suppose $g(x) = x^2$
 then $E(g(X)) = E(X^2)$

Properties:

$$E(aX) = a E(X)$$

$$E(aX + bY) = a E(X) + b E(Y)$$

$$E(X + Y) = E(X) + E(Y)$$

$$E(1) = 1$$

$$E(a) = a$$

$$\begin{aligned} \text{Variance} &= \frac{1}{n} \sum_{i=1}^n x_i^2 p_i - \left(\frac{1}{n} \sum_{i=1}^n x_i p_i \right)^2 = E(X^2) - (E(X))^2 \\ &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 p_i = \sum_{i=1}^n (x_i - E(X))^2 p_i = E(X - E(X))^2 \end{aligned}$$

$$\boxed{\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ \sigma_x^2 &= E[(X - E(X))^2] \end{aligned}}$$

$$SD(X) = \sigma_x = \sqrt{\text{Var}(X)}$$

$$E(g(x)) = \begin{cases} \sum_{i=1}^n g(x_i) \cdot p_i & ; \text{if } X \sim \text{DRV} \\ \int_{-\infty}^{\infty} g(x) \cdot f(x) dx & ; \text{if } X \sim \text{CRV} \end{cases}$$

Choose $g(x) = x$

$$\xrightarrow{\text{Mean of } X} E(X) = \begin{cases} \sum_{i=1}^n x_i \cdot p_i & ; \text{if } X \sim \text{DRV} \\ \int_{-\infty}^{\infty} x \cdot f(x) dx & ; \text{if } X \sim \text{CRV} \end{cases}$$

Choose: $g(x) = X \cdot X = X^2$

$$E(X^2) = \begin{cases} \sum_{i=1}^n x_i^2 \cdot p_i & ; \text{if } X \sim \text{DRV} \\ \int_{-\infty}^{\infty} x^2 \cdot f(x) dx & ; \text{if } X \sim \text{CRV} \end{cases}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= E[(X - E(X))^2] \\ \text{SD} &= \sqrt{\text{Var}(X)} \end{aligned}$$

Mathematical Expectation of One Dimensional Random Variable

Definitions: If X is a discrete RV, then *the expected value* or the mean value of $g(X)$ is defined as

$$E\{g(X)\} = \sum_i g(x_i)p_i,$$

where $p_i = P(X = x_i)$ is the probability mass function of X .

If X is a continuous RV with pdf $f(x)$, then

$$E\{g(X)\} = \int_{R_X} g(x)f(x)dx$$

Mean, Variance and Standard Deviation of a Random Variable

Two expected values which are most commonly used for characterising a RV X are its *mean* μ_X and *variance* σ_X^2 , which are defined as follows:

$$\begin{aligned}\mu_X &= E(X) \\ &= \sum_i x_i P_i, \text{ if } X \text{ is discrete} \\ &= \int_{R_X} x f(x) dx, \text{ if } X \text{ is continuous}\end{aligned}$$

$$\begin{aligned}\text{Var}(X) &= \sigma_X^2 = E\{(X - \mu_X)^2\} \\ &= \sum_i (x_i - \mu_X)^2 p_i, \text{ if } X \text{ is discrete} \\ &\quad \int_{R_X} (x - \mu_X)^2 f(x) dx, \text{ if } X \text{ is continuous}\end{aligned}$$

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2$$

The square root of variance is called *the standard deviation*.

Properties of Expected Value:

1) $E(aX) = a E(X)$

2) $\text{Var}(aX) = a^2 \text{Var}(X)$

3) $E(X+Y) = E(X) + E(Y)$

4) If X and Y are independent

RVs, then $E(XY) = E(X) \cdot E(Y)$

Example:

If the random variable X takes the values 1, 2, 3 and 4 such that $2P(X = 1) = 3P(X = 2) = P(X = 3) = 5P(X = 4)$, find the probability distribution and cumulative distribution function of X .

Solution:

Let $P(X = 3) = 30K$. Since $2P(X = 1) = 30K$, $P(X = 1) = 15K$.

Similarly $P(X = 2) = 10K$ and $P(X = 4) = 6K$.

Since $\sum p_i = 1$, $15K + 10K + 30K + 6K = 1$.

$$\therefore K = \frac{1}{61}$$

The probability distribution of X is given in the following table:

$X = i$	1	2	3	4
P_i	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

The cdf $F(x)$ is defined as $F(x) = P(X \leq x)$. Accordingly the cdf for the above distribution is found out as follows;

When $x < 1$, $F(x) = 0$

When $1 \leq x < 2$, $F(x) = P(X = 1) = \frac{15}{61}$

When $2 \leq x < 3$, $F(x) = P(X = 1) + P(X = 2) = \frac{25}{61}$

When $3 \leq x < 4$, $F(x) = P(X = 1) + P(X = 2) + P(X = 3) = \frac{55}{61}$

When $x \geq 4$, $F(x) = P(x = 1) + P(x = 2) + P(x = 3) + P(x = 4) = 1$.

Example:

A random variable X has the following probability distribution.

$x:$	-2	-1	0	1	2	3
$p(x):$	0.1	K	0.2	$2K$	0.3	$3K$

- (a) Find K , (b) Evaluate $P(X < 2)$ and $P(-2 < X < 2)$, (c) find the cdf of X and (d) evaluate the mean of X .

Solution:

(a) Since $\Sigma P(x) = 1$, $6K + 0.6 = 1$

$$\therefore K = \frac{1}{15}$$

$\therefore$ the probability distribution becomes

x	-2	-1	0	1	2	3
$p(x)$	1/10	1/15	1/5	2/15	3/10	1/5

(b) $P(X < 2) = P(X = -2, -1, 0 \text{ or } 1)$

$$= P(X = -2) + P(X = -1) + P(X = 0) + P(X = 1)$$

[since the events $(X = -2)$, $(X = -1)$ etc. are mutually exclusive]

$$= \frac{1}{10} + \frac{1}{15} + \frac{1}{5} + \frac{2}{15} = \frac{1}{2}$$

$$P(-2 < X < 2) = P(X = -1, 0 \text{ or } 1)$$

$$= P(X = -1) + P(X = 0) + P(X = 1)$$

$$= \frac{1}{15} + \frac{1}{5} + \frac{2}{15} = \frac{2}{5}$$

Solution (Continued):

$$\begin{aligned} \text{(c) } F(x) &= 0, \text{ when } x < -2 \\ &= \frac{1}{10}, \text{ when } -2 \leq x < -1 \\ &= \frac{1}{6}, \text{ when } -1 \leq x < 0 \\ &= \frac{11}{30}, \text{ when } 0 \leq x < 1 \\ &= \frac{1}{2}, \text{ when } 1 \leq x < 2 \\ &= \frac{4}{5}, \text{ when } 2 \leq x < 3 \\ &= 1, \text{ when } 3 \leq x \end{aligned}$$

(d) The mean of X is defined as $E(X) = \sum xp(x)$

$$\begin{aligned} \therefore \text{Mean of } X &= \left(-2 \times \frac{1}{10}\right) + \left(-1 \times \frac{1}{15}\right) + \left(0 \times \frac{1}{5}\right) \\ &\quad + \left(1 \times \frac{2}{15}\right) + \left(2 \times \frac{3}{10}\right) + \left(3 \times \frac{1}{5}\right) \\ &= -\frac{1}{5} - \frac{1}{15} + \frac{2}{15} + \frac{3}{5} + \frac{3}{5} = \frac{16}{15} \end{aligned}$$

Exercise:

Find the value of a , $P(X < 3)$, cumulative distribution function, mean, variance and standard deviation of the discrete random variable (X) with the following probability distribution.

$X = x:$	0	1	2	3	4	5	6	7	8
$p = P(X = x):$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

Exercise:

A random variable X has the following probability distribution.

x	0	1	2	3	4	5	6	7
$p(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

Find (i) the value of K , (ii) $P(1.5 < X < 4.5/X > 2)$

Example:

$$\text{If } p(x) = \begin{cases} x e^{-x^2/2} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- (a) show that $p(x)$ is a pdf (of a continuous RV X) ;
- (b) find its distribution function $P(x)$.

Example:

$$\text{If } p(x) = \begin{cases} x e^{-x^2/2}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- (a) show that $p(x)$ is a pdf (of a continuous RV X).
(b) find its distribution function $F(x)$.

a) $\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 0 dx + \int_0^{\infty} x e^{-x^2/2} dx$

$$I = \int_0^{\infty} x e^{-x^2/2} dx$$

Let $t = x^2 \Rightarrow x = \sqrt{t}$ $\left| \begin{array}{l} x=0 \Rightarrow t=0 \\ x=\infty \Rightarrow t=\infty \end{array} \right.$
 $dt = 2x dx$

$$I = \frac{1}{2} \int_0^{\infty} e^{-t/2} dt$$

$$I = \frac{1}{2} \left[\frac{e^{-t/2}}{-1/2} \right]_0^{\infty} \\ = - \left[0 - 1 \right] \\ = 1$$

b) $F(x) = \int_{-\infty}^x f(x) dx$

$$-\infty < x < 0, F(x) = 0$$

$$0 \leq x < \infty, F(x) = \int_0^x f(x) dx$$

$$= \int_0^x x e^{-x^2/2} dx \quad \boxed{t=x^2}$$

$$F(x) = 1 - e^{-x^2/2}$$

$$F(x) = \begin{cases} 0, & -\infty < x < 0 \\ 1 - e^{-x^2/2}, & 0 \leq x < \infty \end{cases}$$

Solution:

(a) If $p(x)$ is to be a pdf, $p(x) \geq 0$ and

$$\int_{R_X} p(x) \, dx = 1$$

Obviously, $p(x) = xe^{-x^2/2} \geq 0$, when $x \geq 0$

$$\begin{aligned} \text{Now } \int_0^{\infty} p(x) \, dx &= \int_0^{\infty} xe^{-x^2/2} \, dx = \int_0^{\infty} e^{-t} \, dt \quad (\text{putting } t = x^2/2) \\ &= 1 \end{aligned}$$

$\therefore p(x)$ is a legitimate pdf of a RV X .

$$F(x) = P(X \leq x) = \int_0^x f(x) \, dx$$

$\therefore F(x) = 0$, when $x < 0$

$$\text{and } F(x) = \int_0^x xe^{-x^2/2} \, dx = 1 - e^{-x^2/2}, \text{ when } x \geq 0.$$

Example:

If the density function of a continuous RV X is given by

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ 3a - ax, & 2 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

- (i) find the value of a
- (ii) find the cdf of X

Solution:

(i) Since $f(x)$ is a pdf, $\int_{R_x} f(x)dx = 1$

$$\text{i.e.,} \quad \int_0^3 f(x)dx = 1$$

$$\text{i.e.,} \quad \int_0^1 axdx + \int_1^2 adx + \int_2^3 (3a - ax)dx = 1$$

$$\text{i.e.,} \quad 2a = 1$$

$$\therefore \quad a = \frac{1}{2}$$

(ii) $F(x) = P(X \leq x) = 0$, when $x < 0$

$$F(x) = \int_0^x \frac{x}{2} dx = \frac{x^2}{4}, \text{ when } 0 \leq x \leq 1$$

$$= \int_0^1 \frac{x}{2} dx + \int_1^x \frac{1}{2} dx = \frac{x}{2} - \frac{1}{4}, \text{ when } 1 \leq x \leq 2$$

$$= \int_0^1 \frac{x}{2} dx + \int_1^2 \frac{1}{2} dx + \int_2^x \left(\frac{3}{2} - \frac{x}{2} \right) dx = \frac{3}{2}x - \frac{x^2}{4} - \frac{5}{4}, \text{ when } 2 \leq x \leq 3$$
$$= 1, \text{ when } x > 3$$

Example:

A line of length a units is divided into two parts. If the first part is of length X , find $E(X)$, $\text{var}(X)$ and $E\{X(a - X)\}$.

Solution:

Since the positions of the point of division are equally likely, X is uniformly distributed in $(0, a)$.

$$\therefore f(x) = \frac{1}{a}$$

$$E(X) = \int_0^a xf(x) \, dx = \frac{1}{a} \int_0^a x \, dx = \frac{a}{2}$$

$$E(X^2) = \int_0^a x^2 f(x) \, dx = \frac{a^2}{3}$$

$$\therefore \text{Var}(X) = E(X^2) - \{E(X)\}^2 = \frac{a^2}{3} - \frac{a^2}{4} = \frac{a^2}{12}$$

$$E\{X(a - X)\} = a E(X) - E(X^2) = \frac{a^2}{3} - \frac{a^2}{4} = \frac{a^2}{12}$$

Exercise:

A continuous RV X that can assume any value between $x = 2$ and $x = 5$ has a density function given by $f(x) = k(1 + x)$. Find $P(X < 4)$.

Exercise:

A continuous RV X has a pdf $f(x) = kx^2 e^{-x}$; $x > 0$. Find k , mean and variance.

Exercise:

A continuous RV has a pdf $f(x) = 3x^2$; $0 \leq x \leq 1$. Find a and b such that

(i) $P(X \leq a) = P(X > a)$ and

(ii) $P(X > b) = 0.05$

Exercise:

Suppose X is a continuous random variable with the probability density function given

$$\text{by } f(x) = \begin{cases} kx, & 0 \leq x \leq 2; \\ 2k, & 2 \leq x \leq 4, \\ 6k - kx, & 4 \leq x \leq 6. \end{cases}$$

Find the value of k , cumulative distribution function, mean, variance and standard deviation of X .

Two-Dimensional Random Variables

Definitions: Let S be the sample space associated with a random experiment E . Let $X = X(s)$ and $Y = Y(s)$ be two functions each assigning a real number to each outcome $s \in S$. Then (X, Y) is called a *two-dimensional random variable*.

If the possible values of (X, Y) are finite or countably infinite, (X, Y) is called a *two-dimensional discrete RV*. When (X, Y) is a two-dimensional discrete RV the possible values of (X, Y) may be represented as (x_i, y_j) , $i = 1, 2, \dots, m, \dots; j = 1, 2, \dots, n, \dots$.

If (X, Y) can assume all values in a specified region R in the xy -plane, (X, Y) is called a *two-dimensional continuous RV*.

Probability Function of (X, Y)

If (X, Y) is a two-dimensional discrete RV such that $P(x = x_i, y = y_j) = p_{ij}$, then p_{ij} is called the *probability mass function* or simply the *probability function* of (X, Y) provided the following conditions are satisfied.

(i) $p_{ij} \geq 0$, for all i and j

(ii) $\sum_j \sum_i p_{ij} = 1$

The set of triples $\{x_i, y_j, p_{ij}\}$, $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$, ..., is called the *joint probability distribution* of (X, Y) .

Joint Probability Density Function

If (X, Y) is a two-dimensional continuous RV such that

$$P\left\{x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2} \text{ and } y - \frac{dy}{2} \leq Y \leq y + \frac{dy}{2}\right\} = f(x, y) dx dy, \text{ then } f(x, y) \text{ is}$$

called the *joint pdf* of (X, Y) , provided $f(x, y)$ satisfies the following conditions.

(i) $f(x, y) \geq 0$, for all $(x, y) \in R$, where R is the range space.

(ii) $\iint_R f(x, y) dx dy = 1$.

Moreover if D is a subspace of the range space R , $P\{(X, Y) \in D\}$ is defined as

$$P\{(X, Y) \in D\} = \iint_D f(x, y) dx dy. \text{ In particular}$$

$$P\{a \leq X \leq b, c \leq Y \leq d\} = \int_a^b \int_c^d f(x, y) dx dy$$

Cumulative Distribution Function

If (X, Y) is a two-dimensional RV (discrete or continuous), then $F(x, y) = P\{X \leq x \text{ and } Y \leq y\}$ is called *the cdf of (X, Y)* .

In the discrete case,

$$F(x, y) = \sum_j \sum_i p_{ij}$$

$$y_j \leq y, x_i \leq x$$

In the continuous case,

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(x, y) dx dy$$

Properties of $F(x, y)$

- (i) $F(-\infty, y) = 0 = F(x, -\infty)$ and $F(\infty, \infty) = 1$
- (ii) $P\{a < X < b, Y \leq y\} = F(b, y) - F(a, y)$
- (iii) $P\{X \leq x, c < Y < d\} = F(x, d) - F(x, c)$
- (iv) $P\{a < X < b, c < Y < d\} = F(b, d) - F(b, c) + F(a, c) - F(a, d)$
- (v) At points of continuity of $f(x, y)$

$$\frac{\partial^2 F}{\partial x \partial y} = f(x, y)$$

Marginal Probability Distribution

$P(X = x_i) = P\{(X = x_i \text{ and } Y = y_1) \text{ or } (X = x_i \text{ and } Y = y_2) \text{ or etc.}\}$

$$= p_{11} + p_{12} + \dots = \sum_j p_{ij}$$

$P(X = x_i) = \sum_j p_{ij}$ is called the *marginal probability function of X*. It is defined

for $X = x_1, x_2, \dots$ and denoted as $P_{i\cdot}$. The collection of pairs $\{x_i, p_{i\cdot}\}$, $i = 1, 2, 3, \dots$ is called the *marginal probability distribution of X*.

Similarly the collection of pairs $\{y_j, p_{\cdot j}\}$, $j = 1, 2, 3, \dots$ is called the *marginal probability distribution of Y*, where $p_{\cdot j} = \sum_i p_{ij} = P(Y = y_j)$.

In the continuous case,

$$P\left\{x - \frac{1}{2}dx \leq X \leq x + \frac{1}{2}dx, -\infty < Y < \infty\right\}$$

$$= \int_{-\infty}^{\infty} \int_{x - \frac{1}{2}dx}^{x + \frac{1}{2}dx} f(x, y) dx dy$$

$$= \left[\int_{-\infty}^{\infty} f(x, y) dy \right] dx \text{ [since } f(x, y) \text{ may be treated a constant in}$$

$$(x - 1/2dx, x + 1/2 dx)]$$

$$= f_X(x)dx, \text{ say}$$

$f_X(x) = \int_{-\infty}^{\infty} f(x, y)dy$ is called the *marginal density of X*.

Similarly, $f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx$ is called the *marginal density of Y*.

Note

$$P(a \leq X \leq b) = P(a \leq X \leq b, -\infty < Y < \infty)$$

$$= \int_{-\infty}^{\infty} \int_a^b f(x, y) dx dy$$

$$= \int_a^b \left[\int_{-\infty}^{\infty} f(x, y) dy \right] dx = \int_a^b f_X(x) dx$$

Similarly, $P(c \leq Y \leq d) = \int_c^d f_Y(y) dy$

Conditional Probability Distribution

$P\{X = x_i / Y = y_j\} = \frac{P\{X = x_i, Y = y_j\}}{P\{Y = y_j\}} = \frac{p_{ij}}{p_{\bullet j}}$ is called the *conditional probability function of X*, given that $Y = y_j$

The collection of pairs, $\left\{x_i, \frac{p_{ij}}{p_{\bullet j}}\right\} i = 1, 2, 3, \dots,$

is called the *conditional probability distribution of X*, given $Y = y_j$.

Similarly, the collection of pairs, $\left\{Y_j, \frac{p_{ij}}{p_{i\bullet}}\right\}, j = 1, 2, 3, \dots,$ is called the *conditional probability distribution of Y* given $X = x_i$. In the continuous case,

$$\begin{aligned} P\left\{x - \frac{1}{2} dx \leq X < x + \frac{1}{2} dx / Y = y\right\} \\ = P\left\{x - \frac{1}{2} dx \leq X \leq x + \frac{1}{2} dx / y - \frac{1}{2} dy \leq Y \leq y + \frac{1}{2} dy\right\} \\ = \frac{f(x, y) dx dy}{f_Y(y) dy} = \left\{ \frac{f(x, y)}{f_Y(y)} \right\} dx. \end{aligned}$$

$\frac{f(x, y)}{f_Y(y)}$ is called the *conditional density of X*, given Y , and is denoted by $f(x/y)$.

Similarly, $\frac{f(x, y)}{f_X(x)}$ is called the *conditional density of Y*, given X , and is denoted by $f(y/x)$.

Independent RVs

If (X, Y) is a two-dimensional discrete RV such that $P\{X = x_i, Y = y_j\} = P(X = x_i)$

i.e., $\frac{P_{ij}}{P_{\bullet j}} = P_{i\bullet}$, i.e., $P_{ij} = P_{i\bullet} \times P_{\bullet j}$ for all i, j then X and Y are said to be independent RVs.

Similarly if (X, Y) is a two-dimensional continuous RV such that $f(x, y) = f_X(x) \times f_Y(y)$, then X and Y are said to be independent RVs.

Random Vectors

Definitions: A vector $X: [X_1, X_2, \dots, X_n]$ whose components X_i are RVs is called a *random vector*. $(X_1, X_2, \dots, X_n)$ can assume all values in some region R_n of the n -dimensional space. R_n is called the *range space*.

Mathematical Expectation of Two Dimensional Random Variable

Expected Values of a Two-Dimensional RV

If (X, Y) is a two-dimensional discrete RV with joint probability mass function p_{ij} , then $E\{g(X, Y)\} = \sum_j \sum_i g(x_i, y_i) p_{ij}$.

If (X, Y) is a two-dimensional continuous RV with joint pdf $f(x, y)$, then

$$E\{g(X, Y)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) \, dx \, dy$$

Example:

Three balls are drawn at random without replacement from a box containing 2 white, 3 red and 4 black balls. If X denotes the number of white balls drawn and Y denotes the number of red balls drawn, find the joint probability distribution of (X, Y) .

Solution:

As there are only 2 white balls in the box, X can take the values 0, 1 and 2 and Y can take the values 0, 1, 2 and 3.

$$P(X=0, Y=0) = P(\text{drawing 3 balls none of which is white or red})$$

$$= P(\text{all the 3 balls drawn are black})$$

$$= \frac{4C_3}{9C_3} = \frac{1}{21}$$

$$P(X=0, Y=1) = P(\text{drawing 1 red and 2 black balls})$$

$$= \frac{3C_1 \times 4C_2}{9C_3} = \frac{3}{14}$$

$$\text{Similarly, } P(X=0, Y=2) = \frac{3C_2 \times 4C_1}{9C_3} = \frac{1}{7}; P(X=0, Y=3) = \frac{1}{84}$$

$$P(X=1, Y=0) = \frac{1}{7}; P(X=1, Y=1) = \frac{2}{7}; P(X=1, Y=2) = \frac{1}{14};$$

$$P(X=1, Y=3) = 0 \text{ (since only 3 balls are drawn)}$$

$$P(X=2, Y=0) = \frac{1}{21}; P(X=2, Y=1) = \frac{1}{28}; P(X=2, Y=2) = 0;$$

$$P(X=2, Y=3) = 0$$

The joint probability distribution of (X, Y) may be represented in the form of a table as given below:

X	Y			
	0	1	2	3
0	$\frac{1}{21}$	$\frac{3}{14}$	$\frac{1}{7}$	$\frac{1}{84}$
1	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{1}{14}$	0
2	$\frac{1}{21}$	$\frac{1}{28}$	0	0

Example:

For the bivariate probability distribution of (X, Y) given below, find $P(X \leq 1)$, $P(Y \leq 3)$, $P(X \leq 1, Y \leq 3)$, $P(X \leq 1/Y \leq 3)$, $P(Y \leq 3/X \leq 1)$ and $P(X + Y \leq 4)$.

$\begin{array}{c} Y \\ \backslash \\ X \end{array}$	1	2	3	4	5	6
0	0	0	$1/32$	$2/32$	$2/32$	$3/32$
1	$1/16$	$1/16$	$1/8$	$1/8$	$1/8$	$1/8$
2	$1/32$	$1/32$	$1/64$	$1/64$	0	$2/64$

Solution:

$$P(X \leq 1) = P(X=0) + P(X=1)$$

$$= \sum_{j=1}^6 P(X=0, Y=j) + \sum_{j=1}^6 P(X=1, Y=j)$$

$$= \left(0 + 0 + \frac{1}{32} + \frac{2}{32} + \frac{2}{32} + \frac{3}{32} \right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} \right)$$

$$= \frac{1}{4} + \frac{5}{8} = \frac{7}{8}$$

$$P(Y \leq 3) = P(Y=1) + P(Y=2) + P(Y=3)$$

$$= \sum_{i=0}^2 P(X=i, Y=1) + \sum_{i=0}^2 P(X=i, Y=2)$$

$$+ \sum_{i=0}^2 P(X=i, Y=3)$$

$$= \left(0 + \frac{1}{16} + \frac{1}{32} \right) + \left(0 + \frac{1}{16} + \frac{1}{32} \right) + \left(\frac{1}{32} + \frac{1}{8} + \frac{1}{64} \right)$$

$$= \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64}$$

$$P(X \leq 1, Y \leq 3) = \left(0 + 0 + \frac{1}{32} \right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} \right) = \frac{9}{32}$$

$$P(X \leq 1/Y \leq 3) = \frac{P(X \leq 1, Y \leq 3)}{P(Y \leq 3)} = \frac{9/32}{23/64} = \frac{18}{23}$$

$$P(Y \leq 3/X \leq 1) = \frac{P(X \leq 1, Y \leq 3)}{P(X \leq 1)} = \frac{9/32}{7/8} = \frac{9}{28}$$

$$P(X+Y \leq 4) = \sum_{j=1}^4 P(X=0, Y=j) + \sum_{j=1}^3 P(X=1, Y=j) + \sum_{j=1}^2 P(X=2, Y=j)$$

$$= \frac{3}{32} + \frac{1}{4} + \frac{1}{16} = \frac{13}{32}$$

Example:

The joint probability mass function of (X, Y) is given by $p(x, y) = k(2x + 3y)$, $x = 0, 1, 2; y = 1, 2, 3$. Find all the marginal and conditional probability distributions. Also find the probability distribution of $(X + Y)$.

Solution:

The joint probability distribution of (X, Y) is given below. The relevant probabilities have been computed by using the given law.

X	Y		
	1	2	3
0	$3k$	$6k$	$9k$
1	$5k$	$8k$	$11k$
2	$7k$	$10k$	$13k$

$$\sum_{j=1}^3 \sum_{i=0}^2 p(x_i, y_j) = 1$$

i.e., the sum of all the probabilities in the table is equal to 1.

i.e., $72k = 1$.

$$\therefore k = \frac{1}{72}$$

Solution (Continued):

Marginal Probability Distribution of X : $\{i, p_{i*}\}$

$X = i$	$p_{i*} = \sum_{j=1}^3 p_{ij}$
0	$p_{01} + p_{02} + p_{03} = \frac{18}{72}$
1	$p_{11} + p_{12} + p_{13} = \frac{24}{72}$
2	$p_{21} + p_{22} + p_{23} = \frac{30}{72}$
Total = 1	

Marginal Probability Distribution of Y : $\{j, p_{*j}\}$

$Y = j$	$p_{*j} = \sum_{i=0}^2 p_{ij}$
1	$\frac{15}{72}$
2	$\frac{24}{72}$
3	$\frac{33}{72}$
Total = 1	

Solution (Continued):

Conditional distribution of X , given $Y = 1$, is given by $\{i, P(X = i/Y = 1)\} = \{i, P(X = i, Y = 1)/P(Y = 1)\}$
 $= \{i, p_{i1}/p_{*1}\}, i = 0, 1, 2.$

The tabular representation is given below:

$X = i$	p_{i1}/p_{*1}
0	$3k/15k = \frac{1}{5}$
1	$5k/15k = \frac{1}{3}$
2	$7k/15k = \frac{7}{15}$
Total = 1	

The other conditional distributions are given below:

C.P.D. of X , given $Y = 2$	
$X = i$	p_{i2}/p_{*2}
0	$\frac{6k}{24k} = \frac{1}{4}$
1	$\frac{8k}{24k} = \frac{1}{3}$
2	$\frac{10k}{24k} = \frac{5}{12}$
Total = 1	

C.P.D. of X , given $Y = 3$	
$X = i$	p_{i3}/p_{*3}
0	$\frac{9k}{33k} = \frac{3}{11}$
1	$\frac{11k}{33k} = \frac{1}{3}$
2	$\frac{13k}{33k} = \frac{13}{33}$
Total = 1	

Solution (Continued):

C.P.D. of Y, given $X = 0$	
$Y = j$	p_{0j}/p_{0*}
1	$\frac{3k}{18k} = \frac{1}{6}$
2	$\frac{6k}{18k} = \frac{1}{3}$
3	$\frac{9k}{18k} = \frac{1}{2}$
Total = 1	

C.P.D. of Y, given $X = 1$	
$Y = j$	p_{1j}/p_{1*}
1	$\frac{5k}{24k} = \frac{5}{24}$
2	$\frac{8k}{24k} = \frac{1}{3}$
3	$\frac{11k}{24k} = \frac{11}{24}$
Total = 1	

C.P.D. of Y, given $X = 2$	
$Y = j$	p_{2j}/p_{2*}
1	$\frac{7k}{30k} = \frac{7}{30}$
2	$\frac{10k}{30k} = \frac{1}{3}$
3	$\frac{13k}{30k} = \frac{13}{30}$
Total = 1	

Solution (Continued):

Probability distribution of $(X + Y)$	
$(X + Y)$	P
1	$p_{01} = \frac{3}{72}$
2	$p_{02} + p_{11} = \frac{11}{72}$
3	$p_{03} + p_{12} + p_{21} = \frac{24}{72}$
4	$p_{13} + p_{22} = \frac{21}{72}$
5	$p_{23} = \frac{13}{72}$
	Total = 1

Exercise:

If X denotes the number of aces and Y denotes the number of queens obtained when 2 cards are randomly drawn without replacement from a standard deck of cards, then obtain the joint probability distribution of (X, Y) .

Exercise:

The following table represents the joint probability distribution of the two dimensional discrete random variable (X, Y) .

Y	X		
	1	2	3
1	1/12	1/6	0
2	0	1/9	1/5
3	1/18	1/4	2/15

Find all the marginal and conditional probability distributions.

Example:

The joint pdf of a two-dimensional RV (X, Y) is given by $f(x, y) = xy^2 + \frac{x^2}{8}$, $0 \leq x \leq 2, 0 \leq y \leq 1$.

Compute $P(X > 1), P(Y < \frac{1}{2}), P(X > 1/Y < 1/2)$

$P(Y < \frac{1}{2}/X > 1), P(X < Y)$ and $P(X + Y \leq 1)$.

Solution:

Here the rectangle defined by $0 \leq x \leq 2$, $0 \leq y \leq 1$ is the range space R . $R_1, R_2, \dots$, are event spaces.

$$(i) P(X > 1) = \int \int_{R_1} f(x, y) dx dy$$

$$= \int_0^1 \int_1^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy = \frac{19}{24}$$

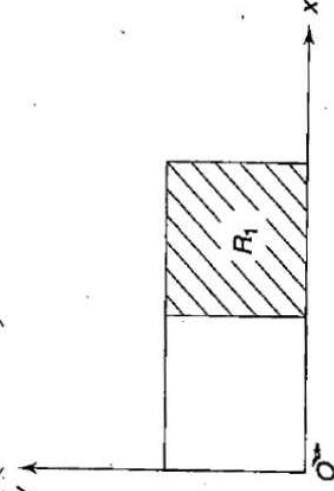


Fig.

$$(ii) P(Y < 1/2) = \int \int_{R_2} \left(xy^2 + \frac{x^2}{8} \right) dx dy$$

$$= \int_0^{1/2} \int_0^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy$$

$$= \frac{1}{4}$$

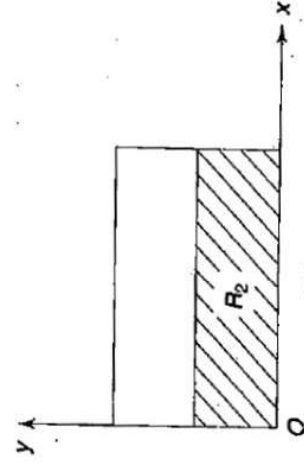


Fig.

Solution (Continued):

$$\begin{aligned}
 \text{(iii) } P(X > 1, Y < 1/2) &= \int_{R_3} \left(xy^2 + \frac{x^2}{8} \right) dx dy \\
 &= \int_0^{1/2} \int_1^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy \\
 &= \frac{5}{24}
 \end{aligned}$$

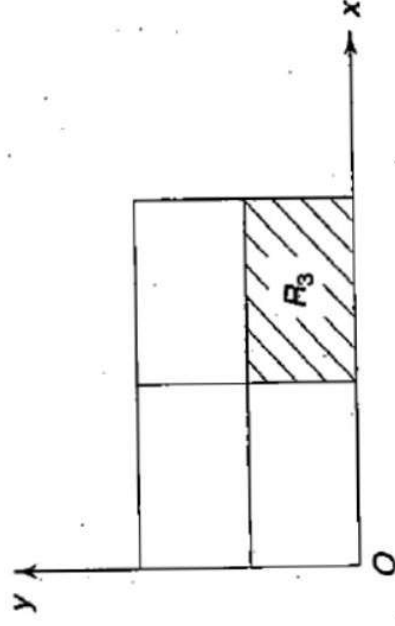


Fig.

$$\text{(iv) } P(X > 1/Y < \frac{1}{2}) = \frac{P\left(X > 1, Y < \frac{1}{2}\right)}{P\left(Y < \frac{1}{2}\right)} = \frac{5/24}{1/4} = \frac{5}{6}$$

$$\text{(v) } P(Y < \frac{1}{2}/X > 1) = \frac{P\left(X > 1, Y < \frac{1}{2}\right)}{P(X > 1)} = \frac{5/24}{19/24} = \frac{5}{19}$$

Solution (Continued):

$$\begin{aligned}
 \text{(vi)} \quad P(X < Y) &= \int_{R_4} \int_{(x < y)} \left(xy^2 + \frac{x^2}{8} \right) dx dy \\
 &= \int_0^1 \int_0^y \left(xy^2 + \frac{x^2}{8} \right) dx dy = \frac{53}{480}
 \end{aligned}$$

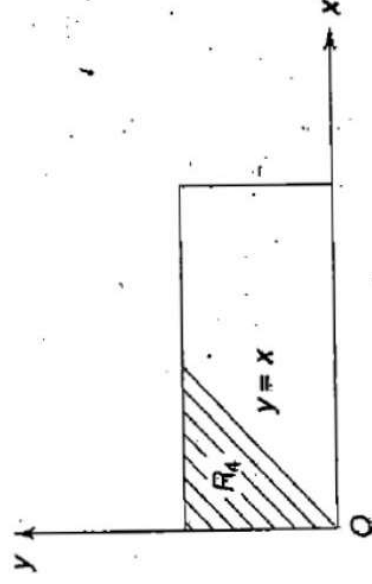


Fig.

$$\begin{aligned}
 \text{(vii)} \quad P(X + Y \leq 1) &= \int_{R_5} \int_{(x+y \leq 1)} \left(xy^2 + \frac{x^2}{8} \right) dx dy \\
 &= \int_0^1 \int_0^{1-y} \left(xy^2 + \frac{x^2}{8} \right) dx dy = \frac{13}{480}
 \end{aligned}$$

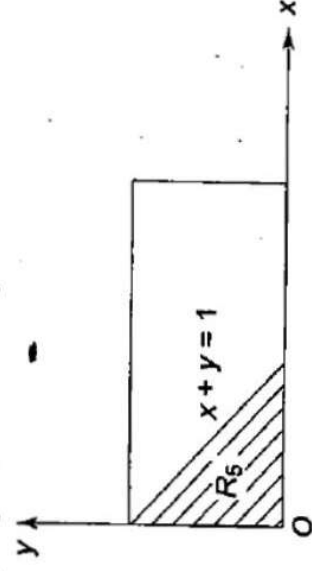


Fig.

Example:

The joint pdf of the RV (X, Y) is given by $f(x, y) = kxy e^{-(x^2 + y^2)} \quad x > 0, y > 0$. Find the value of k and prove also that X and Y are independent.

Solution:

Here the range space is the entire first quadrant of the xy -plane.
By the property of the joint pdf

$$\iint_{x>0, y>0} kxy e^{-(x^2+y^2)} dx dy = 1$$

$$\text{i.e.,} \quad k \int_0^{\infty} ye^{-y^2} dy \int_0^{\infty} xe^{-x^2} dx = 1$$

$$\text{i.e.,} \quad \frac{k}{4} = 1$$

$$\therefore k = 4$$

$$\text{Now } f_X(x) = \int_0^{\infty} 4x e^{-x^2} \times ye^{-y^2} dy = 2x e^{-x^2}, x > 0$$

$$\text{Similarly, } f_Y(y) = 2ye^{-y^2}, y > 0.$$

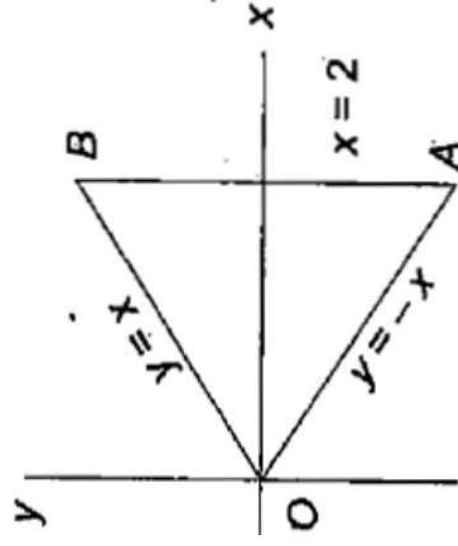
$$\text{Now } f_X(x) \times f_Y(y) = 4x y e^{-(x^2+y^2)} = f(x, y)$$

$\therefore$ The RVs x and y are independent.

Example:

Given $f_{XY}(x, y) = cx(x - y)$, $0 < x < 2$, $-x < y < x$, and 0 elsewhere, (a) evaluate c , (b) find $f_X(x)$, (c) $f_{Y|X}(y/x)$ and (d) $f_Y(y)$.

Solution:



Here the range space is the area within the triangle OAB (shown in the figure), defined by $0 < x < 2$ and $-x < y < x$.

(a) By the property of $jpdf$

$$\int \int_{\Delta OAB} cx(x-y)dx dy = 1$$

$$\int_0^2 \int_{-x}^x cx(x-y)dy dx = 1$$

$$\text{i.e., } 8c = 1$$

$$c = \frac{1}{8}$$

Solution (Continued):

$$(b) f_X(x) = \int_{-x}^x \frac{1}{8} x (x-y) dy$$

$$= \frac{x^3}{4}, \text{ in } 0 < x < 2$$

$$(c) f(y/x) = \frac{f(x,y)}{f_X(x)} = \frac{1}{2x^2} (x-y), -x < y < x$$

$$(d) f_Y(y) = \int_{-y}^2 \frac{1}{8} x (x-y) dx, \text{ in } -2 \leq y \leq 0$$

$$= \int_y^2 \frac{1}{8} x (x-y) dx, \text{ in } 0 \leq y \leq 2$$

$$\text{i.e., } f_Y(y) = \begin{cases} \frac{1}{3} - \frac{y}{4} + \frac{5}{48} y^3, & \text{in } -2 \leq y \leq 0 \\ \frac{1}{3} - \frac{y}{4} + \frac{1}{48} y^3, & \text{in } 0 \leq y \leq 2 \end{cases}$$

Exercise:

If the joint probability density function of a two-dimensional random variable (X, Y) is given by

$$f(x, y) = \begin{cases} k(6 - x - y), & 0 < x < 2, \ 2 < y < 4; \\ 0, & \text{otherwise,} \end{cases}$$

then find (i). the value of k , (ii). $P(Y < 3)$, (iii). $P(X < 1, Y < 3)$, (iv). $P(X + Y < 3)$, and (v). $P(X < 1/Y < 3)$.

Exercise:

If the joint density for the random variables (X, Y) , where X is the unit temperature change and Y is the proportion of spectrum shift that a certain atomic particle produces, is given by

$$f(x, y) = \begin{cases} cxy^2, & 0 < x < y < 1; \\ 0, & \text{otherwise,} \end{cases}$$

then find (i). the value of c , (ii). $E(XY)$, (iii). $f_X(x)$, (iv). $f_Y(y)$ and (v). $f_{Y/X}(y/x)$.

Covariance

As the variance $E\{X - E(X)\}^2$ measures the variations of the R.V. X from its mean value $E(X)$, the quantity $E\{[X - E(X)][Y - E(Y)]\}$ measures the simultaneous variation of two R.V.'s X and Y from their respective means and hence it is called the *covariance* of X , Y and denoted as $\text{Cov}(X, Y)$.

$\text{Cov}(X, Y) = E\{[X - E(X)][Y - E(Y)]\}$ is also called the *product moment* of X and Y and is also denoted as $p(X, Y)$.

$$\text{Cov}(X, Y) = E\{[X - E(X)][Y - E(Y)]\} = E\{XY\} - E(X) \cdot E(Y)$$

$$\text{Cov}(X, Y) = \frac{1}{n} \sum x_i y_i - \frac{1}{n} \sum x_i \cdot \frac{1}{n} \sum y_i$$

Example:

Consider the joint probability distribution

		Y	
		-1	1
X	0	0	$\frac{1}{3}$
	1	$\frac{1}{3}$	$\frac{1}{3}$

Find the co-variance between X and Y .

Solution:

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0.$$

Example:

Let X and Y be continuous random variables with joint pdf

$$f_{X,Y}(x,y) = 3x, \quad 0 \leq y \leq x \leq 1,$$

and zero otherwise. Find the co-variance between X and Y .

Solution:

The marginal pdfs, expectations and variances of X and Y are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy = \int_0^x 3xdy = 3x^2, \quad 0 \leq x \leq 1,$$

$$\Rightarrow E_{f_X}[X] = \int_{-\infty}^{\infty} xf_X(x)dx = \int_0^1 x \times 3x^2dx = \left[\frac{3}{4}x^4\right]_0^1 = \frac{3}{4},$$

$$E_{f_X}[X^2] = \int_{-\infty}^{\infty} x^2f_X(x)dx = \int_0^1 x^2 \times 3x^2dx = \left[\frac{3}{5}x^5\right]_0^1 = \frac{3}{5},$$

$$\Rightarrow Var_{f_X}[X] = E_{f_X}[X^2] - \{E_{f_X}[X]\}^2 = \frac{3}{5} - \left\{\frac{3}{4}\right\}^2 = \frac{3}{80}.$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dx = \int_y^1 3xdx = \left[\frac{3}{2}x^2\right]_y^1 = \frac{3}{2}(1-y^2), \quad 0 \leq y \leq 1,$$

$$\Rightarrow E_{f_Y}[Y] = \int_{-\infty}^{\infty} yf_Y(y)dy = \int_0^1 y \times \frac{3}{2}(1-y^2)dy = \left[\frac{3}{2}\left(\frac{y^2}{2} - \frac{y^4}{4}\right)\right]_0^1 = \frac{3}{2}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{3}{8},$$

$$E_{f_Y}[Y^2] = \int_{-\infty}^{\infty} y^2f_Y(y)dy = \int_0^1 y^2 \times \frac{3}{2}(1-y^2)dy = \left[\frac{3}{2}\left(\frac{y^3}{3} - \frac{y^5}{5}\right)\right]_0^1 = \frac{3}{2}\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{1}{5},$$

$$\Rightarrow Var_{f_Y}[Y] = E_{f_Y}[Y^2] - \{E_{f_Y}[Y]\}^2 = \frac{1}{5} - \left\{\frac{3}{8}\right\}^2 = \frac{19}{320},$$

and to compute the covariance we also need to compute $E_{f_{X,Y}}[XY]$

Solution (Continued):

$$\begin{aligned} E_{f_{X,Y}} [XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x,y) dy dx = \int_0^1 \int_0^x xy \times 3x dy dx \\ &= \int_0^1 \left\{ \int_0^x y dy \right\} 3x^2 dx = \int_0^1 \left[\frac{y^2}{2} \right]_0^x 3x dx = \int_0^1 \frac{x^2}{2} \times 3x^2 dx \\ &= \frac{3}{2} \left[\frac{x^5}{5} \right]_0^1 = \frac{3}{10}, \end{aligned}$$

$$\Rightarrow Cov_{f_{X,Y}} [X, Y] = E_{f_{X,Y}} [XY] - E_{f_X} [X] E_{f_Y} [Y] = \frac{3}{10} - \frac{3}{4} \times \frac{3}{8} = \frac{3}{160}$$

Exercise:

Let X and Y be discrete random variables with joint mass function defined by

$$f_{X,Y}(x,y) = \frac{1}{4}, \quad (x,y) \in \{(0,0), (1,1), (1,-1), (2,0)\},$$

and zero otherwise. Find the co-variance between X and Y .

Exercise:

Two random variables X and Y have the following joint probability density function :

$$f(x, y) = \begin{cases} 2 - x - y; & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ = 0, & \text{otherwise} \end{cases}$$

Find

- (i) Marginal probability density functions of X and Y .
- (ii) Conditional density functions.
- (iii) $\text{Var}(X)$ and $\text{Var}(Y)$.
- (iv) Co-variance between X and Y .

Moment Generating Function (MGF)

Moment Generating Function (MGF) of a RV X (discrete or continuous) is defined as $E(e^{tx})$, where t is a real variable and denoted as $M(t)$.

If X is discrete, then $M(t) = \sum_r e^{tx_r} p_r$,

where X takes the values $x_1, x_2, x_3, \dots$, with probabilities $p_1, p_2, p_3, \dots$

If X is a continuous RV with density function $f(x)$, then

$$M(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

Properties of MGF

(Proofs of the properties are omitted, as the proofs of the corresponding properties of characteristic function will be given later.)

$$1. \quad M(t) = \sum_{n=0}^{\infty} \frac{t^n E(X^n)}{n!}$$

i.e., $E(X^n)$ is the coefficient of $\frac{t^n}{n!}$ in the expansion of $M(t)$ in

series of powers of t .

$$2. \quad \mu_n = E(X^n) = \left[\frac{d^n}{dt^n} M(t) \right]_{t=0}$$

3. If the MGF of X is $M_X(t)$ and if $Y = aX + b$, then $M_Y(t) = e^{bt} M_X(at)$.

4. If X and Y are independent RVs and $Z = X + Y$, then $M_Z(t) = M_X(t) M_Y(t)$.

Example:

If X represents the outcome, when a fair die is tossed, find the MGF of X and hence find $E(X)$ and $\text{Var}(X)$.

Solution:

The probability distribution of X is given by

$$p_i = p(X = i) = \frac{1}{6}, i = 1, 2, \dots, 6$$

$$M(t) = \sum_i e^{tx_i} p_i = \sum_{i=1}^6 e^{ti} p_i$$

$$= \frac{1}{6} (e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t})$$

$$E(X) = [M'(t)]_{t=0} = \frac{7}{2}$$

$$E(X^2) = [M''(t)]_{t=0}$$

$$= \frac{1}{6} [e^t + 4e^{2t} + 9e^{3t} + 16e^{4t} + 25e^{5t} + 36e^{6t}]_{t=0} = \frac{91}{6}$$

$$\text{Var}(x) = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

Example:

If a RV X has the MGF $M(t) = \frac{3}{3-t}$, obtain the standard deviation of X .

Solution:

$$M(t) = \frac{3}{3\left(1 - \frac{t}{3}\right)} = 1 + t/3 + t^2/9 + \dots + \infty$$

$$E(X) = \text{coefficient of } \frac{t}{1} \text{ in } (1) = \frac{1}{3}$$

$$E(X^2) = \text{coefficient of } \frac{t^2}{2} \text{ in } (1) = \frac{2}{9}$$

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2 = \frac{1}{9}$$

$$\sigma_X = \frac{1}{3}$$

Example:

If $x = 1, 2, 3, \dots$ has the geometric distribution $f(x) = pq^{x-1}$, where $q = 1 - p$, show that the moment generating function is

$$M(t) = \frac{pe^t}{1 - qe^t}.$$

Find $E(x)$.

Solution:

The moment generating function of x is

$$\begin{aligned} M(t) &= \sum_{x=1}^{\infty} e^{xt} p q^{x-1} = \frac{p}{q} \sum_{x=1}^{\infty} (q e^t)^x \\ &= p e^t \sum_{x=0}^{\infty} (q e^t)^x = \frac{p e^t}{1 - q e^t}. \end{aligned}$$

To find $E(x)$, we may use the quotient rule to differentiate the expression $M(t)$ with respect to t . This gives

$$\frac{dM(t)}{dt} = \frac{(1 - q e^t) p e^t - p e^t (-q e^t)}{(1 - q e^t)^2}.$$

Setting $t = 0$ gives $E(x) = 1/p$.

Example:

Find the MGF of the binomial distribution and hence find its mean and variance.

Binomial distribution is given by

$$p_r = p(X = r) = {}^nC_r p^r q^{n-r}, \quad r = 0, 1, 2, \dots, n$$
$$p + q = 1$$

Solution:

$$M(t) = \sum_{r=0}^n e^{tr} p_r.$$

$$= \sum_{r=0}^n e^{tr} n C_r p^r q^{n-r}$$

$$= \sum_{r=0}^n n C_r (p e^t)^r q^{n-r}$$

$$= (p e^t + q)^n$$

$$M'(t) = n(p e^t + q)^{n-1} \times p e^t$$

$$M''(t) = np[(p e^t + q)^{n-1} \times e^t + (n-1)(p e^t + q)^{n-2} p e^{2t}]$$

$$E(X) = M'(0) = np \quad (\text{since } p + q = 1)$$

$$E(X^2) = M''(0) = np[1 + (n-1)p]$$

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2$$

$$= np - np^2$$

$$= npq$$

Exercise:

Find the MGF of a RV which is uniformly distributed over $(-1, 2)$ and hence find its mean and variance.

Characteristic Function (CF)

Characteristic function of a RV X (discrete or continuous) is defined as $E(e^{i\omega X})$ and denoted as $\phi(\omega)$.

If X is a discrete RV that can take the values $x_1, x_2, \dots$, such that $P(X = x_r) = p_r$, then

$$\phi(\omega) = \sum_r e^{i\omega x_r} p_r$$

If X is a continuous RV with density function $f(x)$, then

$$\phi(\omega) = \int_{-\infty}^{\infty} e^{i\omega x} f(x) dx$$

Properties of Characteristic Function

1. $\mu'_n = E(X^n)$ = the coefficient of $\frac{i^n \omega^n}{n!}$ in the expansion of $\phi(\omega)$ in series of ascending powers of $i\omega$.

$$2. \mu'_n = \frac{1}{i^n} \left[\frac{d^n}{d\omega^n} \phi(\omega) \right]_{\omega=0}$$

3. If the characteristic function of a RV X is $f_X(\omega)$ and if $Y = aX + b$, then

$$\phi_Y(\omega) = e^{ib\omega} \phi_X(a\omega)$$

4. If X and Y are independent RVs, then

$$\phi_{X+Y}(\omega) = \phi_X(\omega) \times \phi_Y(\omega)$$

Example:

Find the characteristic function of the geometric distribution given by $P(X = r) = q^r p$, $r = 0, 1, 2, \dots, \infty$, $p + q = 1$.

Hence find the mean and variance.

Solution:

$$\phi(\omega) = \sum_{r=0}^{\infty} e^{i\omega r} p q^r$$

$$= p \sum_{r=0}^{\infty} (q e^{i\omega})^r = p(1 - q e^{i\omega})^{-1}$$

$$\phi^{(1)}(\omega) = p(1 - q e^{i\omega})^{-2} i q e^{i\omega}$$

$$\phi^{(2)}(\omega) = i^2 p q [2(1 - q e^{i\omega})^{-3} q e^{i2\omega} + (1 - q e^{i\omega})^{-2} e^{i\omega}]$$

$$E(X) = \frac{1}{i} \phi^{(1)}(0) = \frac{q}{p} \text{ and } E(X^2) = \frac{1}{i^2} \phi^{(2)}(0) = \frac{q}{p^2} (1 + q)$$

$$\mu_X = \frac{q}{p} \text{ and } \sigma_X^2 = \frac{q}{p^2}$$

Exercise:

Find the CF of X whose pdf is given by $f(x) = \frac{1}{2} e^{-|x|}$, $-\infty < x < \infty$.

